

Chinese Remainder Theorem

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$n \geq 2$,

Theorem: For $m_1, \dots, m_n \in \mathbb{N}$, which are mutually coprime, and $a_1, \dots, a_n \in \mathbb{Z}$, there exist a solution (unique modulo $M_1 M_2 \dots M_n$) of the system

$$\begin{cases} x \equiv a_1 \pmod{m_1} \\ x \equiv a_2 \pmod{m_2} \\ \vdots \\ x \equiv a_n \pmod{m_n} \end{cases}$$

Before proving the theorem, introduce a lemma:

Lemma: $\exists$ a solution to the system $\begin{cases} x \equiv 1 \pmod{m_1} \\ x \equiv 0 \pmod{m_2} \end{cases}$,

in which m_1, m_2 are coprime natural numbers.

Proof: let r be inverse of m_2 mod m_1 .

Then for $x = rm_2$, by definition, $x \equiv 1 \pmod{m_1}$,

and since $m_2 \mid x$, $x \equiv 0 \pmod{m_2}$. □

~~From the lemma,~~

Proof of the Theorem

From the lemma, using a similar argument,

$$\exists x' \in \mathbb{Z} \text{ s.t. } \begin{cases} x' \equiv 0 \pmod{m_1} \\ x' \equiv 1 \pmod{m_2} \end{cases}$$

$$\therefore \begin{cases} a_1 x + a_2 x' \equiv a_1 \pmod{m_1} \\ a_1 x + a_2 x' \equiv a_2 \pmod{m_2} \end{cases}$$

Theorem holds when $n = 2$.

To prove the general case, proceed by induction.

~~When~~ ^{if} the theorem holds when $n=k$, then ~~let~~ ^{let} M be ~~the~~ ^a

solution to the system
$$\begin{cases} x \equiv a_1 \pmod{m_1} \\ \vdots \\ x \equiv a_k \pmod{m_k} \end{cases} \Rightarrow$$

$$\therefore x \equiv M \pmod{m_1 \dots m_k}$$

Then for the system
$$\begin{cases} x \equiv a_1 \pmod{m_1} \\ \vdots \\ x \equiv a_{k+1} \pmod{m_{k+1}} \end{cases}$$
 this system can be reduced

to
$$\begin{cases} x \equiv M \pmod{m_1 \dots m_k} \\ x \equiv a_{k+1} \pmod{m_{k+1}} \end{cases}$$
 ← Induction Hypothesis. which has a solution unique up to modulo $m_1 \dots m_{k+1}$ as theorem holds

when $n=2$. Theorem holds for all $n \geq 2$ by induction. □

Example:

$$\begin{cases} x \equiv 2 \pmod{3} \\ x \equiv 4 \pmod{5} \\ x \equiv 6 \pmod{7} \end{cases}, \quad 1000 < x < \del{1200}^{1100}$$

Example:

$$\begin{cases} x \equiv 2 \pmod{3} \\ x \equiv 3 \pmod{5} \\ x \equiv 2 \pmod{7} \end{cases}$$

Solution:

$$x \equiv 35(2)(2) + 21(1)(3) + 15(1)(2) \pmod{105}$$

$$x \equiv 23 \pmod{105}$$

!!

Diophantine Equations

Some equations like $2x + 3y = 1$ have infinitely many INTEGRAL solutions,

but some others like $11x + 22y = 30$ have no solution.

We all know that $11x + 22y = 30$ has no solution because LHS is divisible by 11 and RHS isn't. We thus establish a very simple trick to show an ~~equation~~ equation,

or some subclasses of an equation, has no solution. By modular arithmetic

Unless otherwise specified, we are looking for Integral solutions.

- Examples:
- i) Solve $y^2 = 3x - 1$ mod 3
 - ii) Solve $x^3 + y^3 = 2006$ mod 7
 - iii) Solve $x^2 + y^2 - 5xy = 5$ mod 3 or mod 7.

- Examples:
- i) Solve $y^2 = x^5 - 4$
 - ii) Solve $10^x - 9^y = 1$, $x, y \in \mathbb{N}$.
where $x, y \in \mathbb{N}$
 - iii) Show that $x^2 + y$ & $y^2 + x$ cannot be perfect squares at the same time.

Solutions: i) Consider mod 11.

$$y^2 \equiv 0, 1, 4, 9, 5, 3 \pmod{11}$$

$$x^5 \equiv 0, 1, -1 \pmod{11}$$

$$x^5 - 4 \equiv -4, -3, -5 \pmod{11}$$

$\therefore$ No solution.

Example: i)
$$\begin{cases} x \equiv 1 \pmod{7} \\ x \equiv 4 \pmod{9} \\ x \equiv 3 \pmod{5} \end{cases}$$

ii)
$$\begin{cases} x \equiv 2 \pmod{11} \\ x \equiv 1 \pmod{13} \\ x \equiv 8 \pmod{17} \\ x \equiv 5 \pmod{19} \end{cases}$$

Solution: i)
$$\begin{aligned} x &\equiv 9(5)(5)(1) \\ &\quad + 7(5)(8)(4) \pmod{315} \\ &\quad + 7(9)(2)(3) \\ &\equiv 1723 \pmod{315} \\ &\equiv 148 \pmod{315}. \end{aligned}$$

ii)
$$\begin{aligned} x &= 11a + 2, \text{ some } a \in \mathbb{Z}. \\ 11a + 2 &\equiv 1 \pmod{13}. \\ -2a &\equiv -1 \pmod{13} \\ a &\equiv 7 \pmod{13}. \\ a &= 13b + 7, \text{ some } b \in \mathbb{Z}. \\ x &= 11(13b + 7) + 2 \\ &= 143b + 79. \\ \therefore 143b + 79 &\equiv 8 \pmod{17} \end{aligned}$$

Example: ^{existence} There are n consecutive integers that are not prime powers.

Idea: • Relate with CRT. Need congruence equations to show an integer is NOT a prime power.

- An integer is not a prime power if it has ^{distinct} 2 prime factors $\Rightarrow$ something like $x \equiv 0 \pmod{p_1 p_2}$.
- Consecutive! so consider $x+1 \equiv 0 \pmod{p_1 p_2}$, and so on.

Solution: For distinct primes $p_1, p_2, \dots, p_{n-1}, p_n$.

$$\begin{cases} x \equiv 0 \pmod{p_1 p_2} \\ x \equiv -1 \pmod{p_2 p_3} \\ \vdots \\ x \equiv -(n-1) \pmod{p_{n-1} p_n} \end{cases}$$

$$\therefore \begin{cases} x \equiv 0 \pmod{p_1 p_2} \\ x+1 \equiv 0 \pmod{p_2 p_3} \\ \vdots \\ x+n-1 \equiv 0 \pmod{p_{n-1} p_n} \end{cases}$$

has a solution by CRT.

$x, x+1, \dots, x+n-1$ are not prime powers.

Yay ☺

$$143b \equiv -71 \pmod{17}$$

$$7b \equiv 14 \pmod{17}$$

$$b \equiv 2 \pmod{17}$$

$$b = 17c + 2, \text{ some } c \in \mathbb{Z}.$$

$$\begin{aligned} x &= 143(17c + 2) + 79 \\ &= 2431c + 365. \end{aligned}$$

$$2431c + 365 \equiv 5 \pmod{19}.$$

$$-1(c) \equiv 1 \pmod{19}.$$

$$c \equiv -1 \pmod{19}.$$

Finally, $c = 19d + 18, \text{ some } d \in \mathbb{Z}.$

$$x = 2431(19d + 18) + 365$$

$$= 46189d + 44123.$$

$$\therefore x \equiv 44123 \pmod{46189}$$

ii) $10^x - 9^y = 1, \quad x, y \in \mathbb{N}.$

When $x \geq 2$, $10^x \equiv 0 \pmod{4}$.

However, $9^y \equiv 1 \pmod{4}$, and $0 - 1 \not\equiv 1 \pmod{4}$.

$\therefore x$ cannot be ≥ 2 .

$\therefore x = 1, \quad y = 1.$

Consider mod 8,

$$2^x - 1^y \equiv 1 \pmod{8}.$$

$$2^x \equiv 2 \pmod{8}.$$

$\therefore x = 1, y = 1.$

iii) Assume the contrary. For $x, y \in \mathbb{N}$, s.t. $x^2 + y$ & $y^2 + x$ are both perfect squares.

$$x^2 + y \geq (x+1)^2.$$

$$y \geq 2x+1.$$

Similarly, $y^2 + x \geq (y+1)^2$

$$x \geq 2y+1.$$

But the two inequalities obviously cannot hold at the same time.

~~XX~~ $\square$

Fibonacci Powers

The only Fibonacci powers are 1, 8, 144.

Pythagorean Triples

Propⁿ: For $x, y, z \in \mathbb{N}$, $x^2 + y^2 = z^2$, $(x, y, z) = 1$. Then

$$\begin{cases} x = m^2 - n^2 \\ y = 2mn \\ z = m^2 + n^2 \end{cases}$$

$$(m, n) = 1, m \not\equiv n \pmod{2}, m > n$$

Primitive Pythagorean Triples.

Pell's Equation: $(x^2 - Dy^2 = 1)$ D squarefree

be a minimal solution

Propⁿ: If $(x, y) \in \mathbb{Z}^2$ satisfies $x^2 - Dy^2 = 1$, then all integral solutions of $x^2 - Dy^2 = 1$ are given by $(x, y) = (x_n, y_n)$, where $(a + b\sqrt{D})^n = x_n + y_n\sqrt{D}$.

Example: $(x, y) = (3, 2)$ is a minimal solution to $x^2 - 2y^2 = 1$,
then for $x_n + y_n\sqrt{2} = (3 + 2\sqrt{2})^n$, (x_n, y_n) satisfies $x_n^2 - 2y_n^2 = 1$.

Some Advanced Theorems

Fermat's Last Theorem: For $n \geq 3$, $x^n + y^n = z^n$, where $x, y, z \in \mathbb{Z}$,
we have $xyz = 0$.

Example: $3x^2 + 1 = 4y^3$.

Dirichlet's Theorem: If $(a, b) = 1$, $a, b \in \mathbb{N}$, there are infinitely many primes of the form $ax + b$.

Catalan: The only solution of $x^a - y^b = 1$ in natural numbers for $a, b > 1$ is $(x, a, y, b) = (3, 2, 2, 3)$.